SCIDOFT ACADEMY

OBJECT ORIENTED PROGRAMMING

EPISODE -10

ASSOCIATION, AGGREGATION & COMPOSITION

1. Association
2. Aggregation
3. Composition

How are objects connected?

Start with relationships:

STUDENT ⟷ COURSE

CUSTOMER ⟷ BANK

CAR ⟷ ENGINE

HOUSE ⟷ ROOM

* Objects collaborate with other objects

Association:

* One object is related to or collaborates with another.

TEACHER ⟷ STUDENT

CUSTOMER ⟷ BANK

General Relationship

Aggregation:

Whole ◇—— Part

TEAM ◇—— PLAYER

* Players can conceptually exist independently of a particular team.

HAS-A → with weaker ownership

Composition:

Whole ◇—— PART

HOUSE ◇—— ROOM

* Strong ownership / lifecycle relationship within the model.

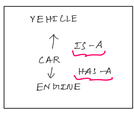
HAS-A → with strong ownership

* The important question is DOMAIN relationship, not merely the symbol.

Remember EPISODE -6 ?

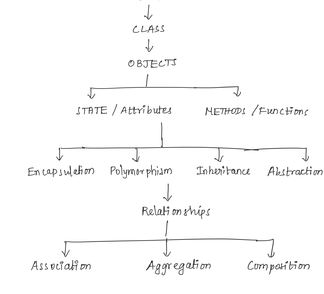
CAR IS-A Vehicle

CAR HAS-A Engine



Don't use inheritance only for code reuse.
Model the actual relationship

PUT EVERYTHING TOGETHER:



THANK YOU!